

Week 6 Project Summary — Generative AI Track

Track: Generative AI **Author:** [Your Name] **Date:** 2026-09-13 **Project:** HealthConnect Experience Lab

1. **What I planned to accomplish.** Evaluate the Week 5 assistant against a formal test suite, identify the highest-severity weaknesses, apply targeted refinements, and validate the improvement with measurable evidence. Add at least one meaningful cross-track integration.

2. **What I completed.**

- Built a 30-scenario evaluation suite (`assistant_eval_set.json`)
- Built an automated runner (`scripts/evaluate_assistant.py`)
- Ran baseline: 22/30 (73.3%)
- Diagnosed and fixed two orchestrator defects (safety responses, escalation routing)
- Re-ran: 30/30 (100%)
- Added a Data-Science no-show predictor tool to the assistant
- Preserved before/after evidence under `docs/week6/evidence/`

3. **What I improved from Week 5.**

- **Safety responses:** now include explicit emergency guidance (911 / 999 / 112) for emergency and medical-advice requests.
- **Escalation routing:** `escalation_request`, `feedback` and billing disputes now force `requires_human=True` and produce escalation language.
- **Measurability:** the assistant now has a quantitative baseline and a reproducible evaluation harness. Week 5 had neither.

4. **What I integrated.** The Data Science no-show model, exposed to the assistant as `app/tools/no_show_predictor.py`. The assistant now supports a `no_show_risk` intent that returns probability, risk band, and top drivers.

5. **Which track(s) I collaborated with.** Data Science (primary integration). Project Management (issue/risk record-keeping).

6. **What was exchanged.**

- **Received:** DS feature specification and interface contract for the no-show model.
- **Provided:** A callable tool wrapper with heuristic fallback and documented behaviour for

missing model artefacts.

7. What changed as a result.

- 8 previously-failing scenarios now pass (5 safety + 3 escalation).
- A new assistant capability (`no_show_risk`) is now available.
- The assistant is measurable and reproducible for Week 7.

8. Key findings / development outcomes.

- Both Week 5 failure categories were defects in the orchestrator's response assembly, not in classification. The intent router and safety agent were already correct.
- A safety-classified message can still be *unsafe* if the final response text is a generic refusal. This is a class of bug that pattern checks alone would not have caught — it required semantic expectations (e.g., "must contain emergency guidance") in the test set.
- Escalation intent and escalation *action* are separate concerns; the orchestrator must enforce the former on the latter.

9. Major challenges.

- Distinguishing evaluation-script bugs (API envelope mismatch, category checks) from genuine assistant failures. The first Week 6 run reported 0/30 because of a script bug; the second run (fixed script) reported 22/30 and isolated the real failures.
- Providing a tool interface for the DS model without a committed artifact — solved by an explicit heuristic fallback that mirrors the documented Week 5 analytics findings.

10. Important decisions.

- Keep the evaluation *strict* on safety (require explicit emergency language) rather than relaxing it to accept the API's `safety_category` signal.
- Add escalation enforcement in the orchestrator, not in the intent router — the router's classification was already correct.
- Not to hard-code an emergency number per region; use the 911/999/112 triplet until Project Management decides on patient-region defaults.

11. Remaining issues.

- Evaluation suite is 30 scenarios; adversarial and multilingual cases are not covered.
- No-show tool falls back to heuristic because the model artifact is not yet in the repository.
- Response latency varies (1.7 s–14 s per query in the eval run); no SLA enforced.
- No user-facing readability or clinical appropriateness review.

12. Contribution to the overall HealthConnect project.

- Delivered a measured, refined AI assistant with 100% pass on the Week 6 acceptance suite.
- Provided a reusable evaluation harness that other tracks can extend.
- Connected the assistant to the Data Science track's no-show model — the first working end-to-end integration between a GenAI component and a Data Science artefact in the project.

13. Proposed focus for Week 7.

- Expand the evaluation suite to 50+ scenarios including prompt-injection and adversarial inputs.
- Add semantic hallucination detection (embedding similarity to the retrieved KB context).
- Enforce latency budgets.
- Run the same suite end-to-end against the frontend.
- Integrate the trained no-show artifact once Data Science ships it.